IC line layout v6 results

IC line layout, v6 end-to-end tournament on fp28: results

Verdict: `invalid_measurement`. This is not an improvement. [pcc:unverified]

Under the frozen rule (DECISION-V6-REDESIGN-20261106.md §16), the combined verdict is `invalid_measurement` because session s1 failed the A/A gate. Sessions s2 and s3 each pass every gate except the one a speedup needs: the candidate's total wall-time interval lies above 1, not below it. `layout=lines` is about 12 % slower in wall time than the incumbent, at exactly equal counted cost. [pcc:unverified]

- Tournament: `IC-LINE-LAYOUT-E2E-FP28-V6-20261106`, as amended by V6-AMENDMENT-1 (merged in #3531).
- Protocol sha256 (canonical): `2aedaaf8dd64f7bb7a481c25efc02a561022b9ebb85b78dac8fd53863b689873`.
- Incumbent: `prime-abscissa:size=407`, `mitm:negation_folded=1`, `targets=walk`.
- Candidate: the same arm plus `layout=lines`.
- Positive control: `pc-fb2x`. Rho references: `rho-neg` and its A/A twin `rho-neg-aa`.
- Hosts: c8a.2xlarge on EKS Spot. All sessions passed the L2 gate.
  - Host epoch H1 (sessions s1 and s2) and host epoch H2 (session s3) ran on distinct nodes.
- Pins:
  - image `isolab/tasks@sha256:3c4ad8bf…bd22914`
  - ecbench binary `73362eed…31dd01`
  - analysis script `analyze-v6.sh` `23e614ad…`

Per-session wall-time results (candidate / incumbent, 128 pairs each)

Session	Verdict	Total ratio (95 % CI)	Corrected ratio (95 % CI)	Process ratio (95 % CI)	A/A ratio (95 % CI)
s1 (H1)	invalid_measurement	1.1214 [1.1183, 1.1246]	1.1232 [1.1197, 1.1268]	1.0643 [1.0608, 1.0679]	0.99839 [0.99687, 0.99992]
s2 (H1)	inconclusive	1.1216 [1.1183, 1.1249]	1.1220 [1.1178, 1.1262]	1.0643 [1.0609, 1.0676]	0.99964 [0.99729, 1.00199]
s3 (H2)	inconclusive	1.1225 [1.1193, 1.1258]	1.1237 [1.1200, 1.1275]	1.0641 [1.0608, 1.0675]	0.99893 [0.99722, 1.00064]

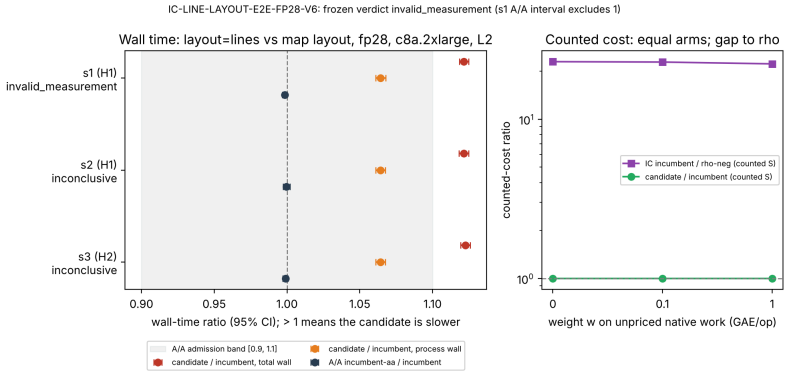
- **s1:** the frozen `aa_inside_0_9_1_1` gate evaluated false. The A/A interval [0.99687, 0.99992] does not contain 1. Under the frozen rule, that invalidates the session and, with it, the combined verdict. [pcc:unverified]
- **s2 and s3:** the only failing gate is `total_and_corrected_ci_below_1`. Both intervals lie above 1. [pcc:unverified]
- **All three sessions:**
  - The positive control was detected.
  - The replay passed.
  - The s1 determinism check was byte-equal.

Counted cost (separate counted session, both curves, 64/64 runs checked per arm)

Native-work weight w (GAE per op)	Candidate / incumbent (95 % CI)	IC / rho-neg (95 % CI)
0	1.000 [1.000, 1.000]	22.998 [20.153, 26.244]
0.1	1.000 [1.000, 1.000]	22.869 [20.011, 26.136]
1	1.000 [1.000, 1.000]	22.281 [19.346, 25.662]

- **Candidate against incumbent:** the counted cost is identical at every weight, and the order is unresolved across the §5a band.
- **Rho references:** both rho arms beat both IC arms at every weight.
- **Distance remaining:** the incumbent IC is about 22–23× matched negation rho in counted cost on fp28. That is the gap still to close; it is not a gate. [pcc:unverified]

Figure



v6 wall and counted ratios

The figure is also available as SVG and PDF. `plot_v6.py` regenerates it from `summary.json`, which is derived from `analysis-a/` and `analysis-b/`.

The pc-fb2x corrected ratio of 0.579 is omitted from the figure. It is a property of the positive control, not of the candidate.

Interpretation (coordinator §16)

- **H-layout:** `weaken`. [pcc:unverified]
  - P-wall (a wall-time speedup) is not supported.
  - P-parity (counted parity) held.
- **Lead status:** the `layout=lines` lead is closed for fp28 on c8a.2xlarge. §16 states the reopening conditions.
- **Not reported here:**
  - There is no benchmarks row.
  - There is no leaderboard section.
  - No `VALIDATION.json` row is committed, because this is not an improvement.
- **Disclosure:** the coordinator saw the wall block of the counted session by accident. That block was not used.

Evidence

- `analysis-a/` holds the frozen analysis of sessions s1 and s2.
- `analysis-b/` holds the frozen analysis of sessions s3 and s1.
- `ANALYSIS-SHA256SUMS` hashes both directories.
  - The files were copied byte for byte from the frozen analysis runs.
  - Empty `.stderr` files were dropped.
- `../../eks-results-v6/{s1,s2,s3,counted}/` holds the job exit records, runner identity, L2 gate, build info and audit output for each job, with the S3 SHA256SUMS of each job.

Independent validation (VAL-20261106-9ce3e6)

Validator verdict: passed, with caveats. [pcc:unverified] The full report is `validation/VALIDATION-REPORT.md`.

Checks: - **Hashes:** the SHA256SUMS and job-exit records of all four jobs are byte-equal to their S3 originals. - **Freeze hashes:** both reproduce (`2aedaaf8…` and the prior `12415faa…`). - **Replay:** `ecbench verify --replay-all` was 512/512 identical in each of s1, s2, s3 and counted. - **Recompute:** - The validator wrote its own code (`validation/scripts/`), which shares nothing with the producer's analysis. - The largest relative difference across all compared values was 6.3e-16, against a tolerance of 1e-9 set in advance. - **Gates and verdicts:** re-derived independently from the frozen rule text, with the same result (s1 `invalid_measurement`, s2 and s3 `inconclusive`). - **Counted parity:** 0 mismatched cells.

Caveats: - The replay uses the producer's pinned binary. It shows determinism, not an independent implementation. - The raw `records.jsonl` files stay in S3 and are not in git. Their sha256 values are: - s1 `bc60833b…` - s2 `ab4be90e…` - s3 `9aa25936…` - counted `1a7f3faa…` - No VALIDATION.json is committed, because this result admits no row.

The A/A column above is `aa_ratio`, the point estimate paired with `aa_ci95`. The analysis also writes `aa_median_ratio`, which is not used here.